

```

echo      7/tcp
echo      7/udp
discard   9/tcp      sink null
discard   9/udp      sink null
sysstat   11/tcp      users
daytime   13/tcp
daytime   13/udp
netstat   15/tcp
qotd      17/tcp      quote
msp       18/tcp      # message send protocol
msp       18/udp
chargen   19/tcp      ttytst source
chargen   19/udp      ttytst source
ftp-data  20/tcp
ftp       21/tcp
fsp       21/udp      fspd
ssh       22/tcp      # SSH Remote Login Protocol
ssh       22/udp
telnet    23/tcp
smtp      25/tcp      mail
time      37/tcp      timeserver
time      37/udp      timeserver
rip       39/udp      resource # resource location
nameserver 42/tcp      name      # IEN 116
whois     43/tcp      nicname
tacacs    49/tcp      # Login Host Protocol (TACACS)

```

Another important file is `/etc/protocols` (you may need to install `nmap` for an extensive list). You can use these files to translate protocol names to numbers (so that the IP layer on other hosts can understand):

```

ip 0 IP # internet protocol, pseudo protocol number
#hopopt 0 HOPOPT # IPv6 Hop-by-Hop Option (RFC1883)
icmp 1 ICMP # Internet control message protocol
igmp 2 IGMP # Internet Group Management
ggp 3 GGP # gateway-gateway protocol
ipencap 4 IP-ENCAP # IP encapsulated in IP (officially "IP")
-----[output truncated]-----

```

The `getprotobyname()` function is the one that gives a `protoent` structure for the line from `/etc/protocols` by matching it with a protocol name. We can show the function as follows:

```

#include <netdb.h>

struct protoent *getprotoent(void);
struct protoent *getprotobyname(const char *name);
struct protoent *getprotobynumber(int proto);
void setprotoent(int stayopen);
void endprotoent(void);

```

The system uses `/etc/nsswitch.conf` to configure which services are to be employed by it in order to find information such as host names, password files, and

group files. Here is a typical entry:

```

passwd:    compat
group:     compat
shadow:    compat

hosts:     files mdns4_minimal [NOTFOUND=return] dns mdns4
networks:  files

protocols: db files
services:  db files
ethers:    db files
rpc:       db files

netgroup:  nis

```

We can use the `ethtool` command followed by the interface name to display the Ethernet card settings:

```

aasisvinayak@GNU-BOK:~$ sudo ethtool eth0
Settings for eth0:

Supported ports: [ TP MII ]
Supported link modes: 10baseT/Half 10baseT/Full
                     100baseT/Half 100baseT/Full
Supports auto-negotiation: Yes
Advertised link modes: 10baseT/Half 10baseT/Full
                     100baseT/Half 100baseT/Full
Advertised auto-negotiation: Yes
Speed: 100Mb/s
Duplex: Full
Port: MII
PHYAD: 0
Transceiver: internal
Auto-negotiation: on
Supports Wake-on: pumbg
Wake-on: g
Current message level: 0x00000033 (51)
Link detected: yes

```

This tool has a wide range of applications—check the man page for details.

We have covered some of the fundamentals in networking, though we still have a lot more basic stuff to discuss—say files like `/etc/inetd.conf`, `/etc/security`, tools like `tcpd` (access control facility), configuration of `/etc/hosts` and so on. We will cover these in the next edition. Then we shall move on to kernel-specific zones. Happy kernel hacking! 

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